W-D-001663(GSA-FSS)
June 15, 1972

INTERIM FEDERAL SPECIFICATION

DISK-PACK, FOR DATA STORAGE
(MAGNETIC RECORDING, 14-INCH, 11 DISKS)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon current available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for magnetic coated 11 disk packs, for use on IBM 2314 and compatible disk drive systems.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

PPP-B-565 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (civil agencies).
Fed. Test Method Std. No. 101B - Preservation, Packaging and Packing Materials: Test Procedures.
Fed. Test Method Std. No. 101/5003 - Compression Test for Shipping Containers.
Fed. Test Method Std. No. 101/5007 - Drop Test (Free Fall).
Fed. Test Method Std. No. 101/5019 - Vibration (Repetitive Shock) Test.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case Waterproof.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 7440

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

National Bureau of Standards:

H-28 - National Bureau of Standards *Handbook of* Screw Thread Standards for Federal Services.

(Applications for copies should be addressed to, Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

Wright Air Development Center, Technical Report:

AD - 286093 - Properties of Glasses at Elevated Temperatures.

(Applications for copies should be addressed to, Clearing House, 5285 Port Royal Road, Springfield, Virginia 22151.)

Uniform Classification Committee:

Uniform Freight Classification.

(Applications for copies should be addressed to, Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 General. Disk packs supplied under this specification shall be new, shall not contain used or re-conditioned parts, and shall have been tested for conformance to the requirements specified herein.

3.2 Product qualification. All magnetic disk packs supplied under this specification shall be products which have been listed on or approved for listing on the applicable Qualified Products List maintained by the General Services Administration (GSA-FSS). For the purpose of product qualification, the term "product" shall include the packaging and packing (see 5.1 and 5.2) as well as the disk pack itself.

3.2.1 Qualified products listing. The Qualified Products List shall consist of products which have been tested and passed those qualification tests specified herein.

3.2.2 Qualification product testing. Products may be submitted by prospective suppliers of magnetic disk packs on dates to be announced by the General Services Administration. Should the submitted product fail to meet specifications as defined herein, the qualifying activity (see 6.5) reserves the right to refuse to accept such products for additional qualification tests until satisfactory data and test results have been submitted indicating correction of product deficiencies. Test results of all products will be available for review to the submitter of the product. The qualifying activity reserves the right to levy a charge to cover cost of testing for qualification of products.

3.2.3 Qualification withdrawal. A supplier's qualified product listing may be withdrawn in accordance with "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPL's)", (see 6.3).

3.2.4 Re-qualification. A supplier's product, once withdrawn from the Qualified Products List, may not be accepted for re-qualification until satisfactory data and test results have been submitted to the qualifying activity (see 6.5) by the supplier indicating correction of the product fault(s).

3.2.5 Process change. Qualification and certification, once established, applies only to those disk packs manufactured by the same process used to make the qualified item. All process changes, including changes in the packaging design or material, shall be reported to the qualifying activity (see 6.5) with a statement by the supplier as to the general nature of the changes and the extent and probable effect of such changes on the delivered product. The qualifying activity reserves the right to require six weeks notice, and suffucient samples of the new product manufactured under any such changes in process. The foregoing requirement does not in any way relieve the manufacturer of delivery requirements, quality control and testing necessary to insure that all products delivered under this specification are equal to those products originally submitted for qualification.

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3.3 Component parts. One self-contained disk pack shall consist of the following list of components (see figure 1).

(a) Top cover (including handle and release spanner).
(b) Bottom cover (including a gasket, bearing, knob and nut or other suitable closure device).
(c) Trim shield.
(d) Disk support and clamping rings.
(e) Top cover disk.
(f) 11 recording disks.
(g) Sector disk.
(h) Spindle locking mechanism (including bushing).
(i) Air filter (including retainer).
(j) Counter balance weight (when necessary).
(k) Hole plugs.
(l) Thrust bearing and washer(s).
(m) Shim ring (when necessary).
(n) Hardware, bolts and nuts, etc.

3.4 Interchangeability. Mechanical, electrical, and functional interchangeability shall exist between disk pack assemblies, subassemblies, and replacement parts. Replacement part interchangeability applies only to parts of a manufacturer's own production.

3.5 Materials. The disk packs may be constructed from any suitable materials as long as the dimensional, inertial, and other functional requirements of this specification are maintained and the finished product does not present an operational hazard.

3.5.1 Toxic compounds. There shall be no component of the disk packs capable of causing bodily harm by contact, inhalation or ingestion during the normal use of the product.

3.5.2 Hardware. The number of different sizes and types of standard hardware items shall be kept to a minimum. Screw threads shall be in accordance with the National Bureau of Standards publication H-28, "Handbook for Screw Thread Standards for Federal Service."

3.5.3 Protective finishes. All material whose function may be adversely affected, or which may adversely affect the function of other components of the disk pack, when exposed to the temperature and humidity conditions specified in 4.3.9.1 shall be suitably protected by a protective finish or coating.

3.5.4 Disk surface coating material. The disk surface coating material shall not unduly suffer impact damage or adversely affect the read/write heads when in normal system use.

3.5.5 Sector disk material. The material for the sector disk shall be an aluminum alloy.

3.5.6 Chemical resistance. The magnetic surface of the recording disks shall be resistant to 91 percent isopropyl alcohol (9 percent water) when used in the cleaning mode (see 6.6.8). No adverse effects on the magnetic or mechanical properties as defined herein shall result from the cleaning.

3.5.7 Glass substrates. Glass disk substrates, in addition to fulfilling all provisions of the specification, shall have a modulus of rupture, when tested in accordance with 4.3.8, of not less than 40,000 pounds per square inch (psi) and the maximum weight of any individual particle obtained upon breakage shall not exceed 0.2 grams.

3.6 Dimensional requirements. When measured in accordance with 4.3.3, the dimensions, tolerances, and physical parameters, shall conform to the values given in figures 2, 3, 4, and 5, and as specified herein.

3.6.1 Top cover outside diameter. The outside diameter of the top cover shall not exceed 14.720 inches. The top cover shall be concentric with the cover centerline within 0.065 inches total indicator reading (see figure 2).

3.6.2 Overall height. The overall height of the disk pack assembly, including the cover set, shall not exceed 8.6 inches (see figure 2).

3.6.3 Overall diameter. The overall diameter of the disk pack assembly, including the cover set, shal. not exceed 15 inches.



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3.6.4 Top cover clearance. When mounted on the disk pack, the vertical clearance between the lower edge of the top cover and the reference line (see figure 2) shall be 0.575 inches ± 0.075 inches.

3.6.5 Disk pack height. The overall height of the disk pack assembly, with cover set removed, shall not exceed 4.535 inches above the reference line (see figure 3).

3.6.6 Hub, spindle lock, and spindle lock shaft dimensions.

3.6.6.1 Hub. The hub shall be constructed so that the bore relief shall conform to figure 3.

3.6.6.2 Spindle lock. The spindle lock thread, chamfer, and spindle lockshaft pin clearance shall conform to figure 3.

3.6.6.3 Locking mechanism clearance. The maximum allowable dimensions for the locking mechanism for the spindle lock shall not exceed the dimensions given in figure 3.

3.6.7 Sector disk. The sector disk shall conform to the dimensions and tolerances given in figure 4.

3.6.8 Hub, disk relationship.

3.6.8.1 Axial runout of disks. A maximum axial runout of 0.012 inches total indicator reading, measured at 13.500 inches ± 0.250 inches diameter, is permitted within the tolerance limits, when the disk pack is rotated at less than 10 rpm.

3.6.8.2 Horizontal runout of disks. The maximum runout for the sector disk and the bottom two magnetic disks shall be 0.010 inches total indicator reading (TIR), and 0.020 inches TIR for the rest of the disks as measured with reference to the disk pack hub centerline.

3.6.9 Disk diameter. The diameters of each recording disk and the top cover disk shall conform to the dimensions and tolerances given in figure 5.

3.6.10 Disk edge chamfer. The disks inner and outer edges may be chamfered as shown in figure 5 or may have any functionally suitable geometry.

3.6.11 Disk support radius. The maximum allowable disk support radius shall be as given in figure 5.

3.6.12 Disk pack clearance. The vertical clearance between the lower most element of the disk pack, with covers removed, and the reference line shall be as given in figure 5.

3.7 Locking force. The disk pack shall be held to the disk drive spindle by a force of 206 pounds ± 32 pounds, exerted by the downward pull of the disk drive lockshaft on the disk pack spindle lock.

3.8 Balance. The disk pack shall be dynamically balanced within 0.14 ounce inches in each of two planes parallel to the disk surface. These planes are located 0.260 inches ± 0.050 inches below and above the top and bottom disk surfaces respectively (see 4.3.4).

3.9 Weight. The weight of the disk pack, with covers, shall not exceed 15 pounds.

3.10 Air feed openings. An opening of 1.1 square inches minimum shall exist in the disk support between each pair of recording surfaces. There shall be a minimum of 12 openings equally spaced around the circumference.

3.11 Air filter. The disk pack shall be provided with an air filter that meets the requirements specified herein.

3.11.1 Filter media. The filter media shall be of screen type construction having a maximum spacing between strands of 0.00173 inches. The filter media shall not be made of a material containing tendrils or loose particles which can separate from the filter and enter the filtered system.

3.11.2 Pressure drop across a clean air filter. The pressure drop across a clean air filter shall not exceed 0.2 inches of water while passing 55 ± 3 standard cubic feet per minute of air.

3.11.3 Air flow through pack. When tested as specified in 4.3.6, the static pressure shall not exceed 0.4 inches of water.



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3.12 Thermal time constant. The disk pack thermal time constant (see 6.6.9) shall not exceed one minute when measured with the disk pack rotating at 2400 ± 48 revolutions per minute in the test environment of 4.3.1.

3.13 Identification. A unique serial number shall be applied to each disk pack. The serial number may be applied by means of a pressure sensitive adhesive label or it may be indelibly stamped. The location of this serial number shall be in an open area and readable without having to remove component parts except for the bottom cover.

3.14 Shock and vibration.

3.14.1 Shock. When tested in accordance with 4.3.7.1, the axial and horizontal runout shall remain in accordance with 3.6.8.1 and 3.6.8.2 and the radial shift of any recording disk shall not exceed 500 microinches.

3.14.2 Vibration. When tested in accordance with 4.3.7.2, all recorded data shall be recovered. No new missing or extra pulse errors (see 6.6.3 and 6.6.4) shall be generated during the tests specified in 4.3.7.1 and 4.3.7.2.

3.15 Magnetic surfaces physical requirements.

3.15.1 Surface roughness. The finished magnetic surfaces shall have a surface roughness of less than 3.5 microinches arithmetic average, with a maximum deviation in height of 30 microinches from the average, when measured using a profile measuring instrument having a 0.0001-inch stylus and a cut-off wave length of 0.030-inches.

3.15.2 Disk pack head gliding requirements. When a disk pack is installed in an IBM 2314, or equivalent, disk drive system, all recording surfaces shall be free of surface irregularities or protrusions that would cause head to disk interference when heads are flown at a spacing of 60 microinches ± 20 microinches.

3.15.3 Dynamic wear. The disk packs shall provide recovery of previously recorded data after 4,000 normal head loadings.

3.16 Magnetic surfaces magnetic requirements.

3.16.1 Magnetic uniformity defects (missing pulse errors and extra pulse errors). When tested in accordance with 4.3.5.2, for missing pulse errors (see 6.6.3) and extra pulse errors (see 6.6.4), there shall be no magnetic uniformity defects in any header area (see 6.6.6) or in track 00000 location, and there shall be a minimum of 4030 tracks free of magnetic uniformity defects.

3.16.2 Resolution ratio. When tested as specified in 4.3.5.3, the ratio of the 2F peak to peak amplitude to the 1F peak to peak amplitude (see 6.6.1 and 6.6.2) shall be 0.50 ± 0.20.

3.17 Environmental requirements.

3.17.1 Accelerated storage life. When tested in accordance with 4.3.9.1, all recorded data shall be successfully recovered, no new missing or extra pulse errors (see 6.6.3 and 6.6.4) shall be generated, and the ability of the disk pack to conform to all other requirements specified herein shall not be impaired.

3.17.2 Disk pack cover set environmental stability. The disk pack cover set shall meet all requirements and dimensions within specified values, after being subjected to the temperature and humidity conditions specified in 4.3.9.2.

3.17.3 Storage life. The storage life of the disk packs (rubber components excluded) shall be five years minimum under temperatures ranging from - 40° F. to 150° F. and relative humidity ranging from 8 to 80 percent.

3.18 Workmanship and cleanliness. The disk packs and containers shall be manufactured and processed in a careful and workmanlike manner in accordance with good practice. Prior to shipping, all disk packs shall be thoroughly cleaned of all flakes, chips, metal particles or other foreign material. There shall be no (as delivered or latent) defects that would adversely effect serviceability, performance, or appearance. The condition of the disk packs and containers shall be examined for workmanship and cleanliness as specified in 4.3.2.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification testing (see 4.2.1).
(b) Quality conformance inspections (see 4.2.2).
(c) Acceptance inspection (see 4.2.3).
(d) Examination of preparation for delivery (see 4.2.4).

4.2.1 Qualification testing. Qualification testing shall be conducted at a laboratory designated by the General Services Administration, Federal Supply Service, and shall consist of all tests listed in table I. All samples subjected to these tests must conform to the associated requirements listed in section 3 of this specification, unless otherwise specified.

4.2.1.1 Samples for qualification. The sample size shall consist of four complete disk packs. All disk packs submitted for qualification testing shall be marked as specified in 3.13. During qualification testing one of the disk packs will be disassembled and portions of the disk pack may be destructively tested.

4.2.1.2 Sample selection. Samples for qualification testing shall be selected by the Government.

4.2.2 Quality conformance inspection. In accordance with 4.1, the contractor shall be responsible for the performance of the quality conformance inspections. The contractor shall select a sample number of disk packs from each lot (see 6.6.7) offered for delivery. Sample disk packs shall be selected in accordance with 4.2.2.2.1 and the contractor shall subject the sample disk packs to the examinations and tests of table I to determine compliance with the specification requirements. All necessary test equipment and facilities required to conduct the subject examinations and tests shall be furnished by the contractor. Alternate test techniques and equipment may be used by the contractor, subject to review and approval by the qualifying activity (see 6.5) that such techniques and equipment are in fact adequate to insure compliance with specification requirements.

4.2.2.1 Reporting of quality conformance test results. Unless otherwise directed by the contracting officer (see 6.1), the contractor shall maintain a complete record of all production test results for the duration of the contract. The production test records shall be available to the Government at all reasonable times. The records shall include the information necessary to identify the lot, individual disk packs, the testing equipment, the inspector, and the date of test. When specified, a copy of all production test results shall be forwarded with each shipment of disk packs to the qualifying activity.

4.2.2.2 Sampling for quality conformance inspection. Sampling procedures for quality conformance inspection shall be in accordance with MIL-STD-105 and as specified herein. The sample disk packs shall be subjected to the Group A and B examinations and tests shown in table I.

4.2.2.2.1 Sampling for group A examinations and tests. The sampling procedure for group A examinations and tests shall be in accordance with MIL-STD-105, general inspection level I at an acceptance quality level (AQL) of 2.5 percent defective.

4.2.2.2.2 Sampling for group B examinations and tests. Group B examinations and tests will be required initially and as often thereafter as is deemed necessary to assure continued compliance with specified requirements. The sampling procedure for group B examinations and tests shall be in accordance with MIL-STD-105, special inspection level S-1 at an acceptance quality level (AQL) of 4.0 percent defective.

4.2.3 Acceptance inspection. Acceptance testing on shipment lots (see 6.6.7) shall be conducted by the qualifying activity (see 6.5). Acceptance tests shall consist of the group A examinations and tests specified in table I.

4.2.3.1 Sampling for acceptance inspection. The sampling procedure for acceptance inspection shall be in accordance with MIL-STD-105, general inspection level I, at an acceptance quality level (AQL) of 2.5 percent defective.



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TABLE I. Inspection Tests

| Test Group | Examinations and Tests | Requirement paragraph | Test paragraph |
|---|---|---|---|
| A | General examination | 3.18 | 4.3.2 |
| | Balance | 3.8 | 4.3.4 |
| | Magnetic uniformity defects | 3.16.1 | 4.3.5.2 |
| B | Dimensional measurements | 3.6 | 4.3.3 |
| | Weight measurement | 3.9 | 4.3.3 |
| | Resolution ratio | 3.16.2 | 4.3.5.3 |
| | Air flow | 3.11.3 | 4.3.6 |
| | Shock | 3.14.1 | 4.3.7.1 |
| | Vibration | 3.14.2 | 4.3.7.2 |
| | Modulus of rupture | 3.5.7 | 4.3.8 |
| | Environmental tests | | |
| | Accelerated storage life | 3.17.1 | 4.3.9.1 |
| | Disk pack cover set | 3.17.2 | 4.3.9.2 |
| | Shipping container tests | | |
| | Vibration | 5.2.1 | 4.3.10.1 |
| | Compression | 5.2.1 | 4.3.10.2 |
| | Drop resistance | 5.2.1 | 4.3.10.3 |

4.2.4 Examination of preparation for delivery. Interior and exterior examinations shall be conducted by a qualifying activity (see 6.5) quality control representative when a lot is offered for acceptance to determine that the packaging, packing, and marking comply with the requirements in section 5, as specified (see 6.1). A sample unit for these examinations shall be one shipping container, fully prepared for delivery. The sample units shall be selected from end item inspection lots and examined for the defects listed in table II. The sampling procedure shall be in accordance with MIL-STD-105, Special Inspection Level S-2 at an acceptance quality level of 6.5 percent defective. The lot size shall be taken as the number of shipping containers in the end item inspection lot.

TABLE II. Preparation for delivery defects

| Examine | Defects |
|---|---|
| Interior packaging | Use of improper or defective materials. |
| | Quantity in unit container not as specified. |
| | Quantity in shipping container not as specified. |
| | Incorrect packaging method employed. |
| | Cushioning or padding omitted. |
| | Cushioning inadequate for the physical protection of the item. |
| Exterior packing and marking | Use of improper or defective materials. |
| | Type, grade, class and style of shipping container not as specified. |
| | Marking incorrect; incomplete, illegible, omitted; of improper size, location, sequence, or method of application. |
| | Inadequate applications of components such as: Incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling. |
| | Bulged or distorted container. |



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4.3 Examinations, tests, and measurements.

4.3.1 Test environment. All tests and measurements required by this specification shall be made at an ambient temperature of 75° F. ± 5° F., and a relative humidity between 45 and 55 percent in a dust free atmosphere. Disk packs to be tested shall be held for a minimum of 24 hours under this test environment before any tests or measurements are made.

4.3.2 General examination. The sample units shall be visually examined to determine conformance to the requirements of 3.13 and 3.18.

4.3.3 Dimensional measurements. The disk packs shall be measured to determine conformance to the dimensional and weight requirements of 3.6 and 3.9.

4.3.4 Balance. The disk packs shall be tested to determine conformance to the balance requirements of 3.8.

4.3.5 Performance tests.

4.3.5.1 Qualification testing equipment. Qualification testing, by the qualifying activity, will be performed using a suitable test processor with associated disk drive unit, precision measuring, balancing, and weighing apparatus and such other equipment deemed necessary to determine conformance of the submitted product to this specification.

4.3.5.1.1 Test heads. Selected data test heads, IBM part numbers 2251097 and 2251098, or equivalent, shall be used for performing magnetic tests.

4.3.5.1.2 Track width tested. The read/write heads on the disk pack drive shall be offset + 0.001 inch and - 0.001 inch from the nominal center line track location, resulting in a track width tested area of 0.009 ± 0.0005 inch.

Note: Nominal track locations shall be established by using an IBM Customer Engineer Disk Pack, Part No. 2200100, or equivalent.

4.3.5.1.3 Write currents. The write current shall be 35 ma ± 1 ma, zero to peak, for writing at track 000 to track 127. The write current shall then be reduced to 30 ma ± 1 ma for writing from track 128 to track 202 (see figure 7).

4.3.5.1.4 Erase current. The erase current, which is a dc current applied to the write coil of the read/write head, shall be of the same magnitude as the write current of 4.3.5.1.3.

4.3.5.1.5 Rotational speed. The maximum rotational speed shall be 2448 revolutions per minute.

4.3.5.2 Magnetic uniformity defects. The disk pack under test shall first be cleaned with 91 percent isopropyl alcohol and a lint free cloth according to the manufacturer's recommendations. Using standard writing procedures, write a 1F signal, with edge erasure, on all track locations and then test for any missing pulse errors. Using standard erase procedures, dc erase all track locations without edge erasure and then test for extra pulse errors. If an error is detected in the header area of a track, this is noted as an error on the first read cycle. The balance of the data area of a track shall be re-read ten times on the detection of an error and if the error re-occurs at least once in the ten re-reads, this shall be recorded as a probable error track. This procedure shall be performed a minimum of three times for all sample disk packs. Any errors that are detected on all three trials shall be recorded as permanent errors. Non-repeatable errors shall be classified as transient errors and not counted. Prior to performing this test the testing equipment will be adjusted and calibrated to the specifications in the applicable reference manuals. The qualifying activity (see 6.5) selected reference disk pack (see 6.4) will be used for calibration purposes. The total number of surface uniformity defects shall not exceed the requirements of 3.16.1.

4.3.5.3 Resolution ratio. Using standard recording procedures, record a track at the 1F frequency, and read back the recorded signals. Note the peak to peak readback amplitude averaged over any 500 microsecond sector. DC erase this track and re-record the same track at the 2F frequency, and read back the recorded signal. Note the peak to peak readback amplitude averaged over the same 500 microsecond sector used for the 1F frequency reading. Calculate the resolution ratio by dividing the 1F peak to peak amplitude by the 2F peak to peak amplitude.

4.3.6 Air flow test. With the disk pack rotating at 2400 ± 48 revolutions per minute and with an air flow, measured at the air intake, through the disk pack of 53 ± 3 standard cubic feet per minute, the static pressure shall be measured between the atmosphere and the static pressure tap for conformance to 3.11.3. The test configuration and location of measuring points are given in figure 6.



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4.3.7 Shock and vibration tests.

4.3.7.1 Shock. After noting the number and location of all magnetic uniformity defects, the disk pack shall be recorded on all tracks at the LF frequency (see 6.6.1) and subjected to three exposures of each of the conditions listed in table I. After completion of the shock tests, the protective covers shall not be cracked, broken, or rendered unserviceable. The axial and horizontal runout of each disk shall be measured to determine conformance with 3.6.8.1 and 3.6.8.2. The radial shift of any recording disk shall not exceed the value specified in 3.14.1.

4.3.7.2 Vibration. On the satisfactory completion of the test in 4.3.7.1, the disk pack with covers shall then be attached to the vibrating table with two tie-down straps. Each tie-down strap shall be a 0.375-inch diameter shock cord or suitable equivalent. The straps shall pass through the disk pack handle and the attachment of the four strap ends to the vibrating table shall form a 13.5-inch square pattern. The resulting total tie-down force (perpendicular to disk surfaces) shall be 45 ± 5 pounds. The assembly shall be subjected to simple harmonic motion having a constant peak to peak displacement of 0.1 inches ± 10 percent in the frequency range from 5 to 20 Hz, and a constant peak to peak acceleration of 20 units (772 inches/second$^2$) ± 10 percent in the frequency range from 20 to 1000 Hz. The frequency range of 5 to 1000 Hz shall be transversed at a logarithmic rate in 5 minutes ± 10 percent. Vibration shall continue for 2 hours with direction of motion perpendicular to the disk surfaces and for an additional 2 hours with directions of motion parallel to the disk surfaces. The disk pack shall then be cleaned if necessary and shall be placed on the same drive used to record it. The disk pack shall be read back, erased without edge erasure, and re-read to determine if recorded data has been lost and to determine the presence and location of any new, missing or extra pulse errors for conformance to 3.14.2.

4.3.8 Modulus of rupture (glass substrates). The modulus of rupture shall be determined by a four point loading fixture test using a procedure equivalent to that described in Wright Air Development Center Technical Report, "Properties of Glasses at Elevated Temperatures," Document No. AD-286093 for conformance to the requirement of 3.5.7.

4.3.9 Environmental tests.

4.3.9.1 Accelerated storage life. One disk pack shall be selected at random noting the number and location of all magnetic uniformity defects. The disk pack shall be recorded on all tracks at the LF frequency (see 6.6.1) and, with covers removed, placed in a temperature and humidity controlled chamber and held for five full temperature and humidity cycles as shown in figure 8. The disk pack shall then be removed from the chamber, shall be held for a period of 24 hours in the environment specified in 4.3.1, and shall then be given visual, balance, and dimensional examinations for conformance to requirements specified herein. The disk pack shall then be cleaned if necessary and placed on the drive used to record it. The disk pack shall be read back, erased without edge erasure, and re-read to determine if recorded data has been lost and to determine the presence and location of any new, missing or extra pulse errors for conformance to 3.17.1.

4.3.9.2 Disk pack cover set. One disk pack with covers attached shall be subjected to the test conditions specified in 4.3.9.1 (simultaneously with that test) except that only the cover set shall be examined, both visually and dimensionally, for conformance with 3.17.2.

4.3.10 Shipping container tests. Not less than two qualification test samples shall be subjected to the following cycle of tests before being removed from their shipping containers.

4.3.10.1 Vibration (repetitive shock). The shipping containers to be tested shall be subjected to the vibration test defined by Method 5019 of Fed. Test Method Std. 101B. (DO not open the container until 4.3.10.3 has been completed.)

4.3.10.2 Compression. The shipping containers to be tested shall be subjected to a momentary top-to-bottom load to 500 pounds using test method 5003 of Fed. Test Method Std. No. 101B. This method is the same as ASTM method D 642-47.

4.3.10.3 Drop resistance (free fall). The shipping containers to be tested shall be subjected to the drop test defined by Method 5007 of Fed. Test Method Std. No. 101B. The 30 inch drop height shall be used, and procedure A/26 (one of each face, edge and corner) shall be employed.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.1).

5.1.1 Level A. The disk packs shall be individually packaged in close-fitting heat sealed bags. The bags shall be constructed from film conforming to L-P-378, type I, grade A, finish 1. The material shall be a minimum of .002 inches thick.



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5.1.2 Level B. The disk packs shall be individually packaged as specified in 5.1.1 except that the bag may be closed with a wire or clip of the type normally used for closing this type of bag.

5.1.3 Level C. The disk packs shall be individually packaged in accordance with the manufacturer's standard practice providing that this will insure protection for the disk pack during shipment and provides for safe delivery to its destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Each disk pack packaged as specified in 5.1 shall be cushioned on its top, bottom and periphery with an appropriate nondusting cushioning material and enclosed in a close-fitting box conforming to PPP-B-636, type CF, class weather-resistant, variety DW. Close, waterproof seal, and strap the boxes in accordance with the appendix of PPP-B-636. The filled, closed, and sealed containers shall be capable of withstanding the cycle of tests prescribed in 4.3.10 in such a manner that the disk pack contained therein shall meet all of the functional requirements of this specification.

5.2.2 Level B. Each disk pack packaged as specified in 5.1 shall be packed and tested as specified in 5.2.1 except that the box shall conform to PPP-B-636, type CF, class domestic, variety SW or DW, grade 275 minimum.

5.2.3 Level C. Each disk pack packaged as specified in 5.1 shall be packed to assure carrier acceptance and safe arrival at destination in compliance with the National Motor Freight Classification rules and the Uniform Freight Classification rules. When fiberboard boxes are used, the fiberboard from which they are fabricated shall conform to PPP-F-320.

5.3 Marking. In addition to the markings required by 5.3.3, all marking shall be in accordance with 5.3.1 or 5.3.2.

5.3.1 Civil agencies. In addition to the markings required by 5.3.3, all marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to the markings required by 5.3.3, all marking shall be in accordance with MIL-STD-129.

5.3.3 Special marking. In addition to any special marking required by the contract or order, and the marking required by 5.3.1 or 5.3.2, all shipping containers shall be marked with an identification as Disk Pack, 11-High, part number, lot number, date of manufacture, contract number, and Federal Stock number, if available.

6. NOTES

6.1 Ordering data. Purchasers should exercise the preferred options permitted herein and include the following information in procurement documents:

(a) Title, symbol and date of this specification.
(b) Quantity of disk packs required.
(c) Inspection responsibility, if other than specified (4.1).
(d) Copies of production test results, if required (4.2.2.1).
(e) Test record retention period, if other than specified (4.5.1.2).
(f) Special marking, if required.
(g) Level of packaging and packing required (5.1 and 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-W-D-001663 whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Federal Supply Service, General Services Administration, Washington, D. C. 20406. Application for qualification tests shall be made in accordance with "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPL's)", (see 6.3).

6.3 Copies of "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPL's)" may be obtained upon application to Director, Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C 20406.



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6.4 The qualifying activity (see 6.5) will provide output calibration of a prospective supplier's reference disk pack, upon request. Requests of this nature should be directed to Director, Magnetic Surfaces Laboratory (GSA-FSS), Gaithersburg, Maryland (located on the grounds of the National Bureau of Standards, Gaithersburg, Maryland).

6.5 Qualifying activity. The term qualifying activity as used herein shall mean the General Services Administration.

6.6 Definitions:

6.6.1 LF frequency. The LF frequency is 2.5 mega flux reversals per second (mfrps).

6.6.2 2F frequency. The 2F frequency is 5.0 mega flux reversals per second (mfrps).

6.6.3 Missing pulse error. A missing pulse error is either a single pulse or a series of consecutive pulses of a LF frequency signal (see 6.6.1), that has a base to peak amplitude of less than 25 percent of the track 200 reference (peak to peak) amplitude (see 6.6.5).

6.6.4 Extra pulse error. An extra pulse error is any spurious signal, on a track that has been previously recorded at the LF frequency (see 6.6.1) and then de erased, with a base to peak amplitude greater than 22.5 percent of the track 200 reference (peak to peak) amplitude (see 6.6.5).

6.6.5 Reference amplitude. The reference amplitude is defined as that peak to peak amplitude resulting when the qualifying activity (see 6.5) selected reference pack is recorded at the LF frequency (see 6.1.1) using the write currents specified herein (see 4.3.5.1.3).

6.6.6 Header area. The header area is defined as beginning 100 microseconds ± 5 microseconds after index point and ending 700 microseconds ± 5 microseconds after index point with the disk pack rotating at 2400 rpm.

6.6.7 Inspection lot. A lot shall consist of enough disk packs manufactured from a common lot of materials and selected from concurrent production to meet the requirements of the contract or purchase order.

6.6.8 Cleaning code. A lint free swab shall be moistened with the isopropyl alcohol and rubbed lightly against the rotating disk surface for the purpose of removing foreign material from the surface. Isopropyl alcohol shall be reagent grade and the water (s - 3.5.6) shall be deionized or distilled. Proportions shall be by volume.

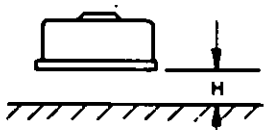
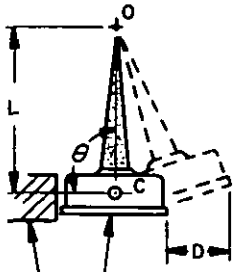
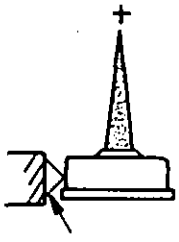
6.6.9 Thermal time constant. The thermal time constant is defined as that time required for the disk pack to reach two-thirds of the original drive/pack temperature differential.



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TABLE I

TEST CONDITIONS FOR SHOCK TESTS
OF DISK PACK AND COVER ASSEMBLY

| TEST | SCHEMATIC | COND. A WITH TOP AND BOTTOM COVERS ON 1/ | COND. B WITH BOTTOM COVER REMOVED 1/ |
|---|---|---|---|
| FREE FALL (FLAT ± 2°)<br>STEP 1 |  <br>HARD SMOOTH RIGID SURFACE (SUCH AS CONCRETE FLOOR) | H= 4 in. | H= 2.5 in. |
| SIDE IMPACT (FLAT SURFACE)<br>STEP 2 |  <br>LIGHTWEIGHT RIGID SUSPENSION WHICH ROTATES FREELY ABOUT PIVOT "O" BUT PREVENTS ROTATION ABOUT AXIS O-C AND MAINTAINS θ = 90°<br>PACK C.C.<br>FIXED RIGID FLAT IMPACT SURFACE 2/ | L = 30 in.<br>D= 8 in. | D= 6 in. |
| SIDE IMPACT (SHARP CORNER)<br>STEP 3 |  <br>SAME TEST SETUP AS ABOVE<br>RIGID FIXED SQUARE BASED PYRAMID HAVING 90° APEX WITH A RADIUS OF 1/16 INCH 2/ | D= 6 in.<br>IMPACT AGAINST EDGE OF 7TH DATA DISK FROM TOP (i.e., DATA SURFACE 11 AND 12) | D= 5 in.<br>IMPACT AGAINST EDGE OF BOTTOM INDEX DISK |

1/ TOLERANCE ON ALL DIMENSIONS IS ± 1/8 INCH UNLESS OTHERWISE SPECIFIED
2/ RECOMMEND USE OF MASSIVE STEEL OR ALUMINUM IMPACT ELEMENTS.



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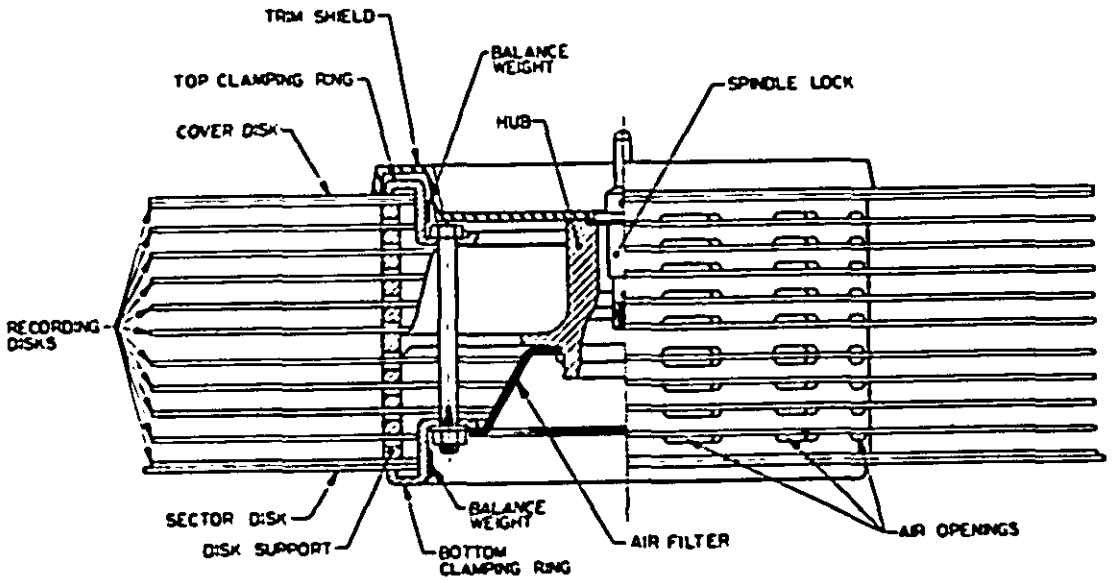
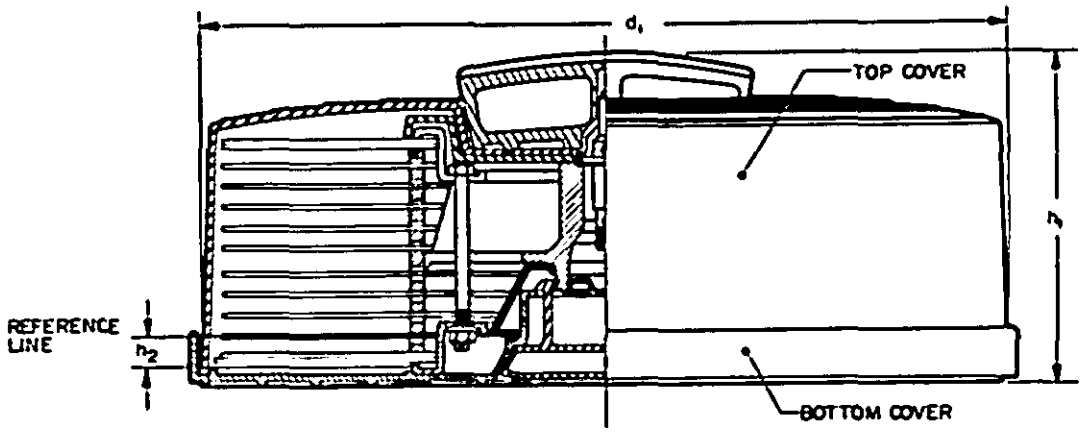



FIGURE 1

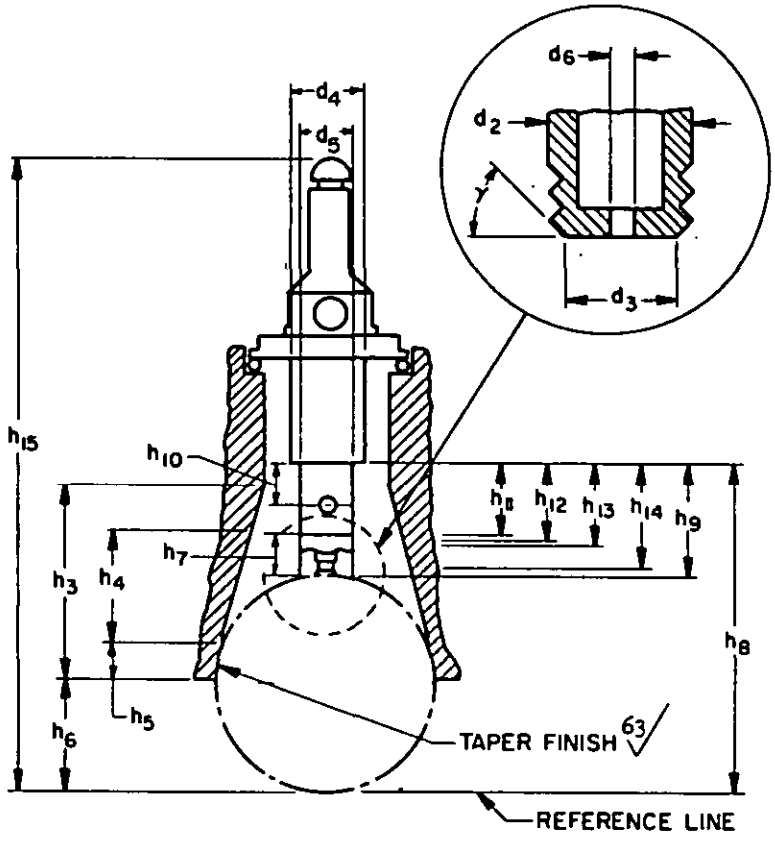



| DIMENSION | MEASUREMENT |
|---|---|
| $h_1$ | 6.600 IN. MAX. |
| $h_2$ | 0.575±0.075 IN. |
| $d_1$ | 14.720 IN. MAX. |

FIGURE 2



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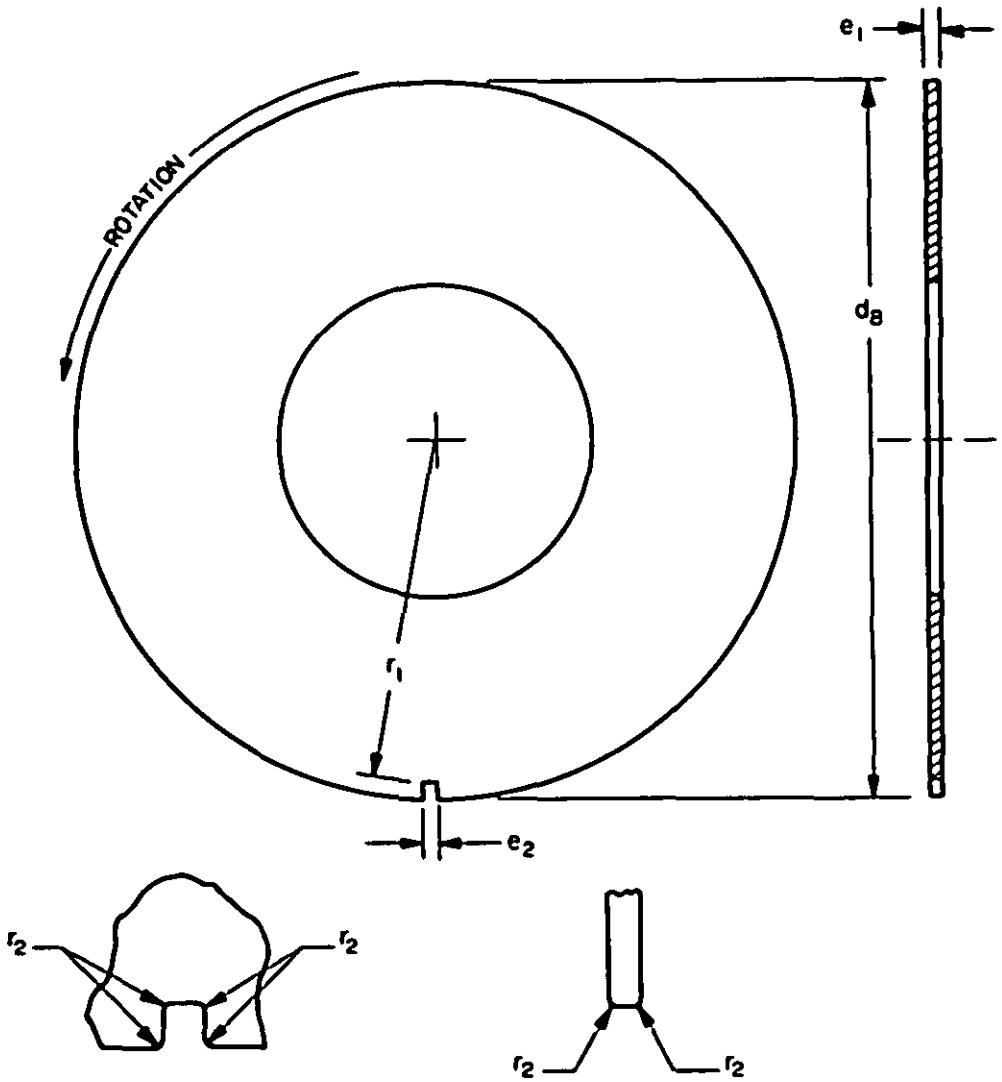

| DIMENSION | MEASUREMENT |
|---|---|
| $h_3$ | 1.390 ± 0.015 IN. |
| $h_4$ | 0.810 ± 0.060 IN. |
| $h_5$ | 0.250 ± 0.030 IN. |
| $h_6$ | 0.822 ± 0.012 IN. |
| $h_7$ | 0.290 IN. MINIMUM |
| $h_8$ | 2.325 ± 0.022 IN. |
| $h_9$ | 0.774 ± 0.006 IN. |
| $h_{10}$ | 0.341 ± 0.009 IN. |
| $h_{11}$ | 0.492 IN. MAXIMUM |
| $h_{12}$ | 0.543 ± 0.020 IN. |
| $h_{13}$ | 0.563 IN. MINIMUM |
| $h_{14}$ | 0.728 IN. MAXIMUM |
| $h_{15}$ | 4.535 IN. MAXIMUM |
| $d_2$ | 0.375 IN. NOMINAL |
| $d_3$ | 0.290 ± 0.012 IN |
| $d_4$ | $0.432\ ^{+0.000}_{-0.005}$ IN. |
| $d_5$ | 0.380 IN. MAXIMUM |
| $d_6$ | 0.125 IN. MINIMUM |
| $\gamma$ | 30° TO 45° ± 2° |
| TAPER | (1±0.0002):2 IN. |

NOTE: When measured from the spindle lock shoulder: $h_{10}$ = distance to safety balls; $h_{11}$ = distance to threaded portion of spindle lock; $h_{12}$ = clearance for lockshaft pin penetration; $h_{13}$ = lockshaft penetration depth at which safety balls cease to expand; and $h_{14}$ = lockshaft penetration depth at which top cover must be removable.

FIGURE 3



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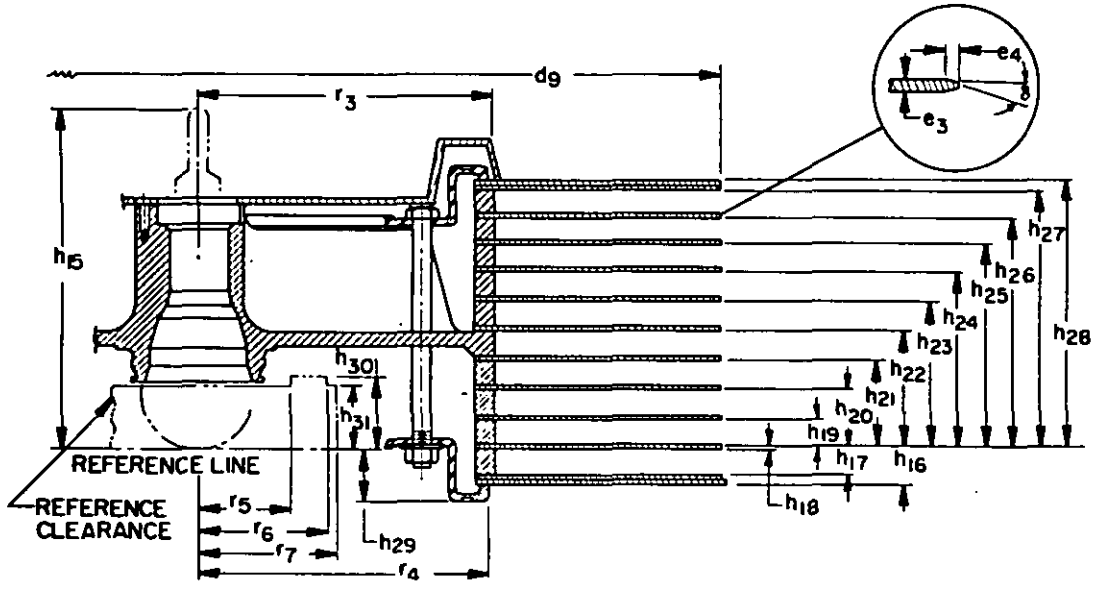



| DIMENSION | MEASUREMENT |
|---|---|
| $d_8$ | 14.281 ± 0.010 IN. |
| $r_1$ | 7.000 ± 0.008 IN. |
| $r_2$ | 0.015 IN MAXIMUM |
| $e_1$ | 0.051 ± 0.003 IN. |
| $e_2$ | 0.075 ± 0.005 IN. |

FIGURE 4



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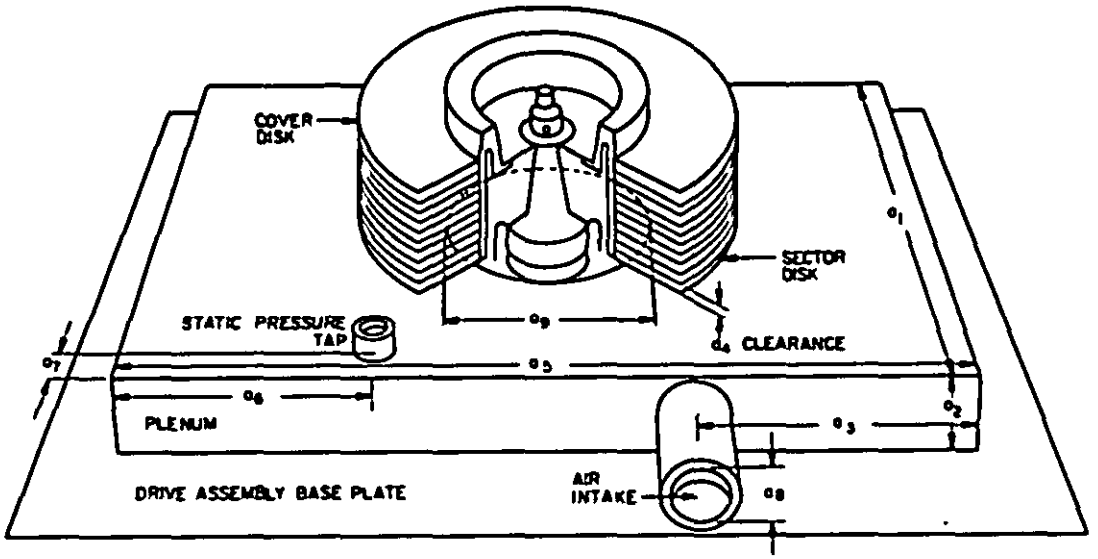



| DIMENSION | MEASUREMENT |
|---|---|
| $d_9$ | 14.025 ± 0.005 IN. |
| $r_3$ | 3.950 IN. MAXIMUM |
| $r_4$ | 3.543 IN. MAXIMUM |
| $r_5$ | 1.280 IN. |
| $r_6$ | 1.590 IN. |
| $r_7$ | 1.660 IN. |
| $h_{15}$ | 4.535 IN. MAXIMUM |
| $h_{16}$ | 0.481 IN. $^{+0.020}_{-0.025}$ IN. |
| $h_{17}$ | 0.380 ± 0.016 IN. |
| $h_{18}$ | 0.030 ± 0.016 IN. |
| $h_{19}$ | 0.370 ± 0.016 IN. |
| $h_{20}$ | 0.770 ± 0.016 IN. |
| $h_{21}$ | 1.170 ± 0.016 IN. |
| $h_{22}$ | 1.570 ± 0.016 IN. |
| $h_{23}$ | 1.970 ± 0.016 IN. |
| $h_{24}$ | 2.370 ± 0.016 IN. |
| $h_{25}$ | 2.770 ± 0.016 IN. |
| $h_{26}$ | 3.170 ± 0.016 IN. |
| $h_{27}$ | 3.570 ± 0.016 IN. |
| $h_{28}$ | 3.670 ± 0.019 IN. |
| $h_{29}$ | 0.725 IN. MAXIMUM |
| $h_{30}$ | 0.980 IN. MINIMUM |
| $h_{31}$ | 0.790 IN. MINIMUM |
| $e_3$ | 0.050 ± 0.001 IN. |
| $e_4$ | 0.050 IN. MAXIMUM |
| $\delta$ | 15° ± 2° |

FIGURE 5



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| DIMENSION | MEASUREMENT |
|---|---|
| $a_1$ | 15.6 IN. |
| $a_2$ | 2.5 IN. |
| $a_3$ | 10.4 IN. |
| $a_4$ | .13 IN. |
| $a_5$ | 31.5 IN. |
| $a_6$ | 9.4 IN. |
| $a_7$ | 1.2 IN. |
| $a_8$ | 2.0 IN. |
| $a_9$ | 7.2 IN. |

NOTES

1. PLENUM COMPLETELY SEALED EXCEPT FOR AIR INTAKE & $a_8$ HOLE
2. PLENUM PRESSURE CONTROLLED BY THROTTLING AIR INTAKE

FIGURE 6

| ZONE A | ZONE B |
|---|---|
| (TRACK 000-127) | (TRACK 128-202) |
| $I_W$ = 35 MA ± 1 MA | $I_W$ = 30 MA ± 1 MA |
| $I_O$ = .08 $I_W$ MAXIMUM | $I_O$ = .08 $I_W$ MAXIMUM |
| $T_R$ AND $T_F$ = 140ns ± 40ns | $T_R$ AND $T_F$ = 140ns ± 40ns |
| $T_R$ = $T_F$ ± 20ns | $T_R$ = $T_F$ ± 20ns |
| $T_1$ = $T_2$ ± 2 PERCENT | $T_1$ = $T_2$ ± 2 PERCENT |

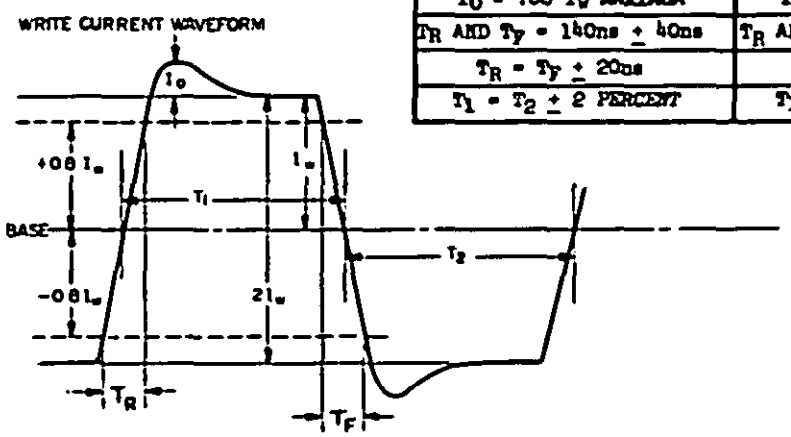
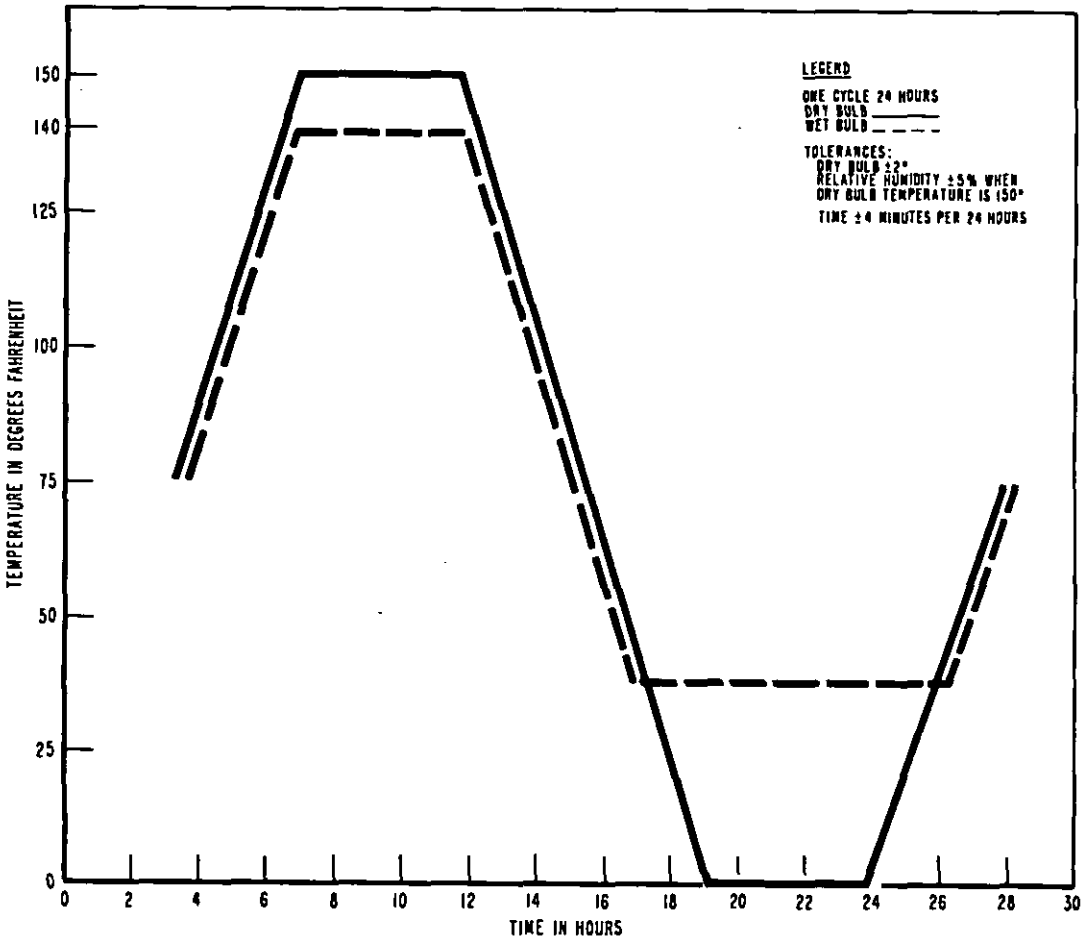



FIGURE 7



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TEMPERATURE - HUMIDITY CYCLE
ENVIRONMENTAL TEST
FIGURE 8.



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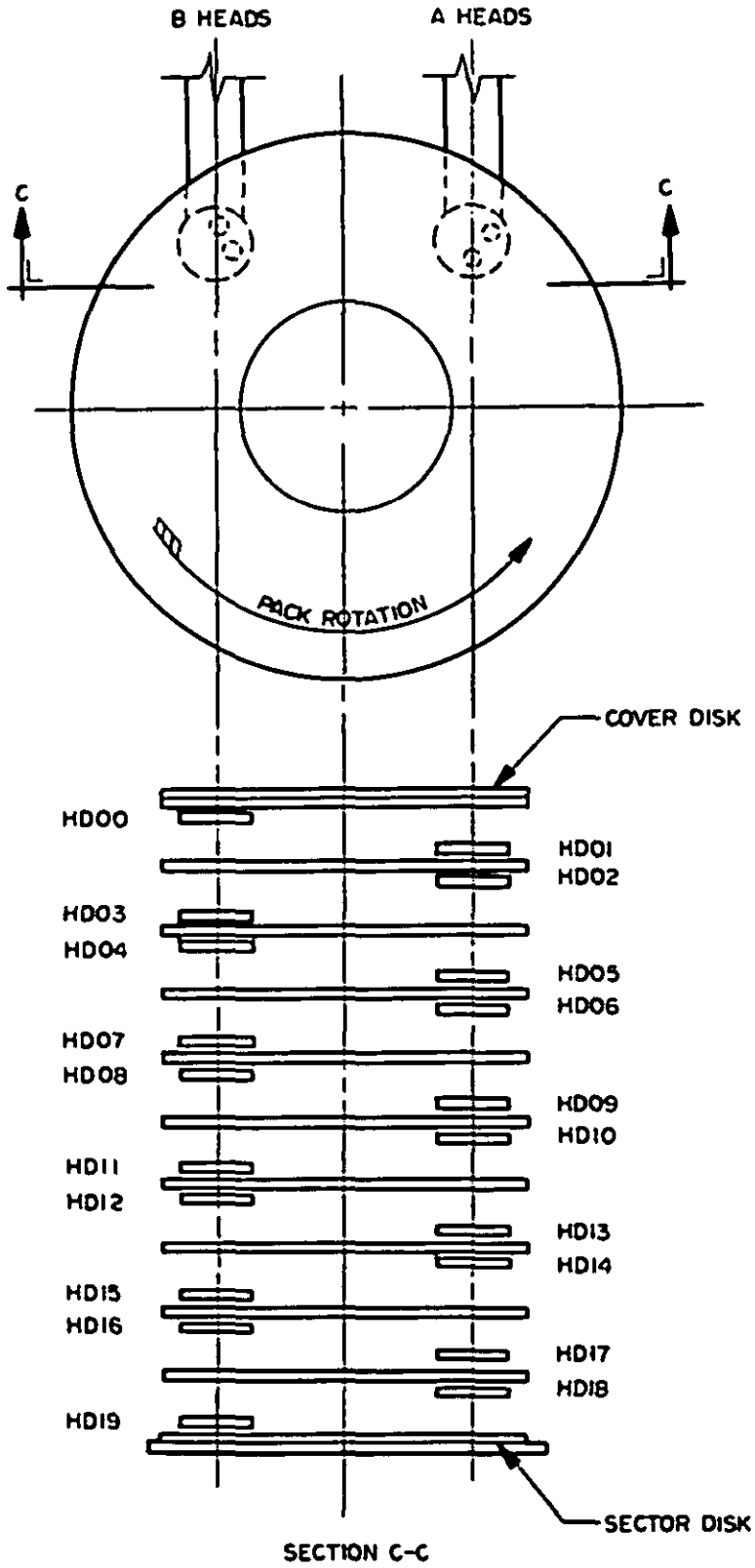



FIGURE 9. DISK ROTATION, DISK SURFACE AND HEAD DESIGNATION



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